

Q42

$$\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt{n} e^n}$$

$\therefore$ Converges by DLT.

$$\begin{aligned} \frac{\ln n}{\sqrt{n} e^n} &\leq \frac{1}{e^n} = \left(\frac{1}{e}\right)^n \\ \sum \left(\frac{1}{e}\right)^n &\text{converges} \end{aligned}$$